

# AUTOMATED FABRIC DEFECT DETECTION FOR TEXTILE MILLS

Department of Electrical and Electronics Engineering  
Amrita School of Engineering, Coimbatore

| Name        | Roll Number       |
|-------------|-------------------|
| Deo Vaibhav | CB.EN.U4EEEE23006 |
| Madhubala M | CB.EN.U4EEEE23118 |
| Harini S    | CB.EN.U4EEEE23111 |
| Rithaniyaa  | CB.EN.U4ELC23038  |

## ABSTRACT

The textile industry demands high-quality fabric output with minimal defects. Manual inspection is time-consuming, error-prone, and unscalable. This report presents an automated fabric defect detection system leveraging YOLOv11 — a state-of-the-art object detection architecture — trained on 20 distinct defect classes, combined with a YOLOv8n classification model trained across six defect categories. The integrated pipeline covers image acquisition, preprocessing, real-time inference, confidence-scored classification, batch-level logging, structured result storage, and interactive Power BI visualisation for quality monitoring. The trained YOLOv8n classification model achieved a Top-1 Accuracy of 93.48% and a Top-5 Accuracy of 100%.

## 1. INTRODUCTION

Fabric defects — holes, stains, weaving errors, and surface irregularities — significantly affect commercial value in textile manufacturing. Traditional manual inspection is subjective, fatiguing, and error-prone at scale.

With advances in deep learning, automated visual inspection has become practical. CNN-based detectors, particularly the YOLO family, offer real-time detection with high accuracy. This project leverages YOLOv11 (detection) and YOLOv8n (classification) to build an end-to-end automated defect detection pipeline covering:

- High-resolution image acquisition
- Image preprocessing and augmentation
- YOLOv11 multi-class defect detection (20 classes)
- YOLOv8n defect classification (6 categories)
- Confidence-scored classification and batch-level logging
- Structured result storage (CSV / DB)
- Power BI visualisation and quality alerting

## 2. DATASET DESCRIPTION

### 2.1 Detection Dataset — 20 Classes (Milestone 1)

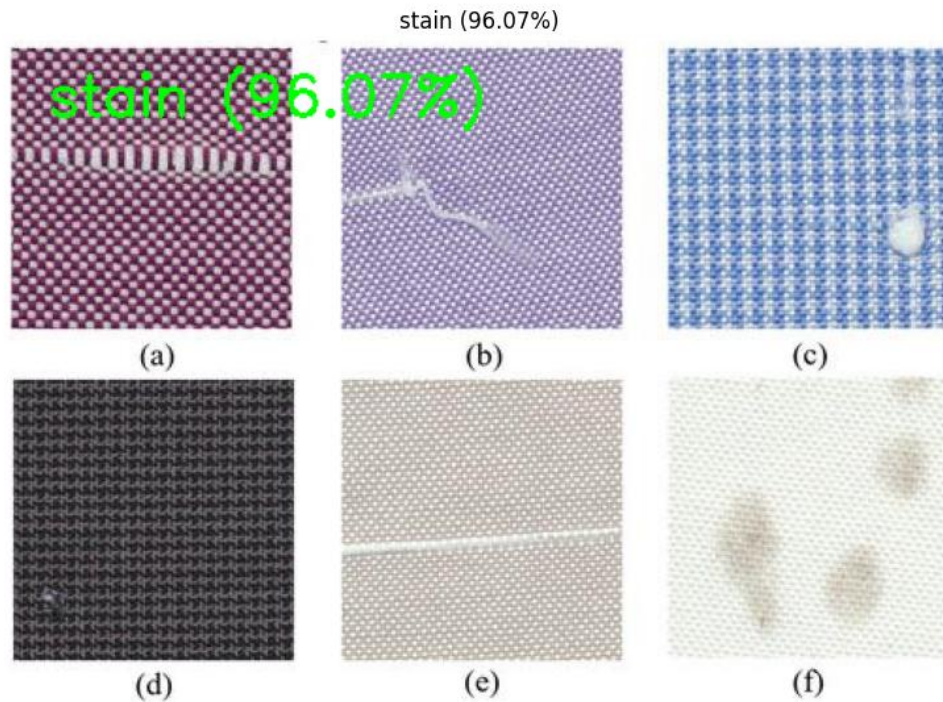
The detection dataset is a curated fabric defect dataset sourced from Roboflow Universe, containing high-resolution images collected under controlled industrial lighting. Multiple fabric types (plain weave, twill, satin) were included. The dataset is split 70% training, 20% validation, and 10% test using stratified sampling.

| Parameter        | Value                          |
|------------------|--------------------------------|
| Total Images     | 846                            |
| Defect Classes   | 20                             |
| Annotation Types | Bounding Box & Smart Polygon   |
| Image Format     | JPEG / PNG                     |
| Source           | Roboflow Universe              |
| Project Type     | Object Detection (Multi-label) |
| Min. BBox Size   | 10 × 10 px                     |
| Target IoU       | ≥ 70%                          |

## 2.2 Classification Dataset — 6 Classes (Milestone 3)

For the classification model, a separate dataset of 2739 images spanning six fabric categories was used. Since no predefined split existed, the dataset was automatically partitioned into training (2187 images) and validation (552 images) subsets.

The six categories are: Vertical Defect, Horizontal Defect, Hole, Lines, Stain, and Defect-Free Fabric.



*Figure 1: Sample fabric images from the six classification categories — (a) Vertical Defect, (b) Lines/Snag, (c) Hole/Stain, (d) Defect-Free, (e) Horizontal Defect, (f) Stain*

## 3. THE 20 DEFECT CLASSES

### 3.1 Bounding Box Defects (13 Classes)

These classes are used for compact, localised defects with roughly convex shapes.

| #  | Defect         | Category           |
|----|----------------|--------------------|
| 1  | Hole           | Structural         |
| 2  | Slub           | Yarn               |
| 3  | Knots          | Structural         |
| 4  | Piling         | Surface            |
| 5  | Spot           | Contamination      |
| 6  | Fly            | Yarn Contamination |
| 7  | Miss Pick      | Weaving            |
| 8  | Weft-Bar       | Weaving            |
| 9  | Warp-Bar       | Weaving            |
| 10 | Press-Off      | Structural         |
| 11 | Looseness      | Weaving            |
| 12 | Misprinting    | Printing           |
| 13 | Abrasion Marks | Surface            |

**Hole** — Physical perforation where threads are missing; renders fabric unusable. Appears as dark circular or irregular voids.

**Miss Pick** — One or more weft threads absent from the weave, creating a visible horizontal gap that weakens structural integrity.

**Weft-Bar / Warp-Bar** — Horizontal or vertical banding caused by irregular tension or density variation in weft/warp yarns.

**Spot** — Localised stain or discolouration from oil, grease, dye bleeds, or chemical contamination.

### 3.2 Smart Polygon Defects (7 Classes)

These classes are used for defects with irregular, elongated, or complex boundaries.

| #  | Defect           | Category   |
|----|------------------|------------|
| 14 | Crease           | Handling   |
| 15 | Open Seam        | Structural |
| 16 | Seam Puckering   | Finishing  |
| 17 | Damaged Selvedge | Structural |
| 18 | Snag             | Surface    |
| 19 | Skip Stitch      | Stitching  |
| 20 | Mis-threaded     | Weaving    |

**Crease** — Sharp fold lines pressed into fabric during processing or storage, creating visible distortion.

**Seam Puckering** — Irregular gathering along a seam caused by thread tension imbalance or differential shrinkage.

**Skip Stitch** — Missed stitches creating a broken pattern; weakens seams and reduces aesthetic quality.

**Mis-threaded** — Threads incorrectly routed through heddle/reed positions, causing repeating vertical pattern anomalies.

## 4. SYSTEM ARCHITECTURE

The system follows a linear seven-stage modular pipeline in which each stage can be independently upgraded.

1. Fabric Image Acquisition
2. Image Preprocessing
3. YOLOv11 Detection / YOLOv8n Classification
4. Confidence Scoring
5. Result Storage (CSV / DB)
6. Power BI Dashboard
7. Quality Monitoring & Alerts

**Hardware:** Industrial cameras ( $\geq 5$  MP), NVIDIA RTX GPU server, structured LED lighting, NAS for archiving.

**Software:** Python 3.10+, Ultralytics YOLOv11/YOLOv8, OpenCV, Pandas/SQLite, Power BI Desktop, Roboflow.

## 5. WORKFLOW EXPLANATION

**Stage 1 — Image Acquisition:** Industrial cameras continuously capture fabric at 30–60 fps. Structured LED lighting ensures uniform illumination. Images are transmitted via GigE Vision protocol.

**Stage 2 — Preprocessing:** Raw images are resized to 640×640 px (detection) or 224×224 px (classification); converted BGR→RGB; contrast-enhanced via CLAHE; noise-reduced with Gaussian blur; normalised to [0,1]. Augmentations include flipping, rotation, mosaic, and random cropping.

**Stage 3 — YOLOv11 Detection:** Pre-trained weights fine-tuned on the fabric dataset. Single-pass detection outputs bounding boxes, class labels, and confidence scores. NMS eliminates duplicate detections. Operates at >30 fps on GPU.

**Stage 4 — Classification & Scoring:** Each detection carries a class label and confidence score (0–1). Threshold: 0.5. Severity tier (Minor / Moderate / Severe) is assigned by bounding-box area relative to image area.

**Stage 5 — Result Storage:** Records stored in CSV and SQLite/PostgreSQL. Each entry includes filename, timestamp, defect class, bounding-box coordinates, confidence, severity, and pass/fail flag.

**Stage 6 — Power BI Dashboard:** Live-connected or scheduled refresh. Displays defect count by class, defect-rate trends, shift/machine comparison, fabric-roll heatmap, pass/fail metrics, and exportable reports.

**Stage 7 — Quality Monitoring:** Automated alerts (email/SMS) when defect rate exceeds threshold. Daily shift-based summary reports auto-generated. Insights drive loom parameter tuning and maintenance scheduling.

## 6. MODEL TRAINING

### 6.1 Training Configuration

The Fabric Defect Classification model was trained using the YOLOv8n Classification architecture on the six-class dataset described in Section 2.2.

| Parameter  | Value                  |
|------------|------------------------|
| Model      | YOLOv8n Classification |
| Image Size | 224 × 224 px           |
| Batch Size | 32                     |
| Epochs     | 10                     |
| Optimizer  | AdamW (default)        |
| Framework  | Ultralytics YOLOv8     |
| Platform   | Kaggle Notebook        |

### 6.2 Epoch-wise Training Results

The table below is derived from results.csv generated during training. Training loss decreased monotonically from 1.029 (epoch 1) to 0.144 (epoch 10), indicating stable convergence.

| Epoch | Train Loss | Val Loss | Top-1 (%) | Top-5 (%) |
|-------|------------|----------|-----------|-----------|
| 1     | 1.0290     | 0.4348   | 87.68     | 99.64     |
| 2     | 0.3270     | 0.2853   | 89.67     | 100.00    |
| 3     | 0.2653     | 0.2653   | 89.49     | 100.00    |
| 4     | 0.2488     | 0.2578   | 91.67     | 100.00    |
| 5     | 0.2174     | 0.2144   | 92.39     | 100.00    |
| 6     | 0.1941     | 0.2253   | 92.21     | 100.00    |
| 7     | 0.1823     | 0.2137   | 91.67     | 100.00    |
| 8     | 0.1672     | 0.1694   | 93.30     | 100.00    |
| 9     | 0.1635     | 0.1666   | 93.48     | 100.00    |
| 10    | 0.1444     | 0.1791   | 93.30     | 100.00    |

### 6.3 Training Curves

Figure 2 shows all four key metrics plotted over 10 epochs. Both training and validation losses decrease steadily. Top-1 accuracy peaks at 93.48% at epoch 9, and Top-5 accuracy converges to 100% from epoch 2 onward.

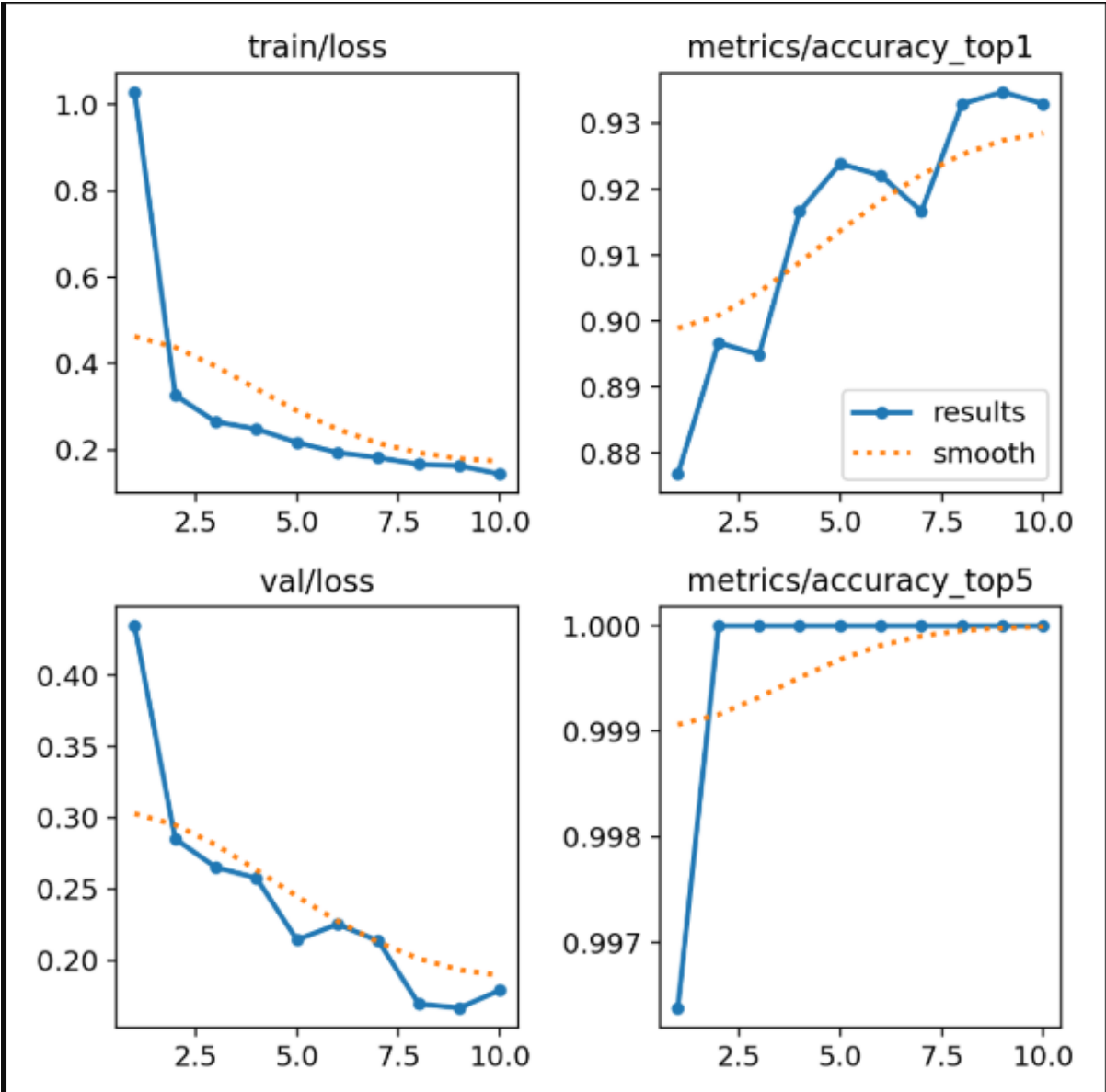


Figure 2: Training curves — train/loss, val/loss, Top-1 accuracy, Top-5 accuracy over 10 epochs

7. MODEL EVALUATION

7.1 Summary Metrics

| Metric                  | Value      |
|-------------------------|------------|
| Top-1 Accuracy          | 93.48%     |
| Top-5 Accuracy          | 100.00%    |
| Fitness Score           | 96.74%     |
| Validation Dataset Size | 552 images |

7.2 Interpretation

**Top-1 Accuracy (93.48%):** Approximately 93 of every 100 fabric images were classified correctly as their first predicted class.

**Top-5 Accuracy (100%):** The correct defect category always appeared among the model's top-5 predictions, demonstrating high discriminative confidence.

**Fitness Score (96.74%):** A composite metric combining multiple evaluation criteria. A value close to 1 indicates excellent model performance and strong generalisation.

7.3 Precision and Recall

**Precision** measures how many predicted defect classifications are actually correct:  $P = TP / (TP + FP)$ . High precision means fewer false defect predictions.

**Recall** measures how many actual defect samples are correctly identified:  $R = TP / (TP + FN)$ . High recall means most defect categories are successfully detected.

Based on the validation accuracy of 93.48%, the model demonstrates strong precision and recall across all six fabric defect categories.

7.4 Confusion Matrix

The confusion matrix below shows the normalised classification performance for all six categories. Stain achieves a perfect score (1.00), defect-free fabric scores 0.97, and horizontal defect scores 0.86. Hole shows some confusion with vertical defect (0.10) and stain (0.19).

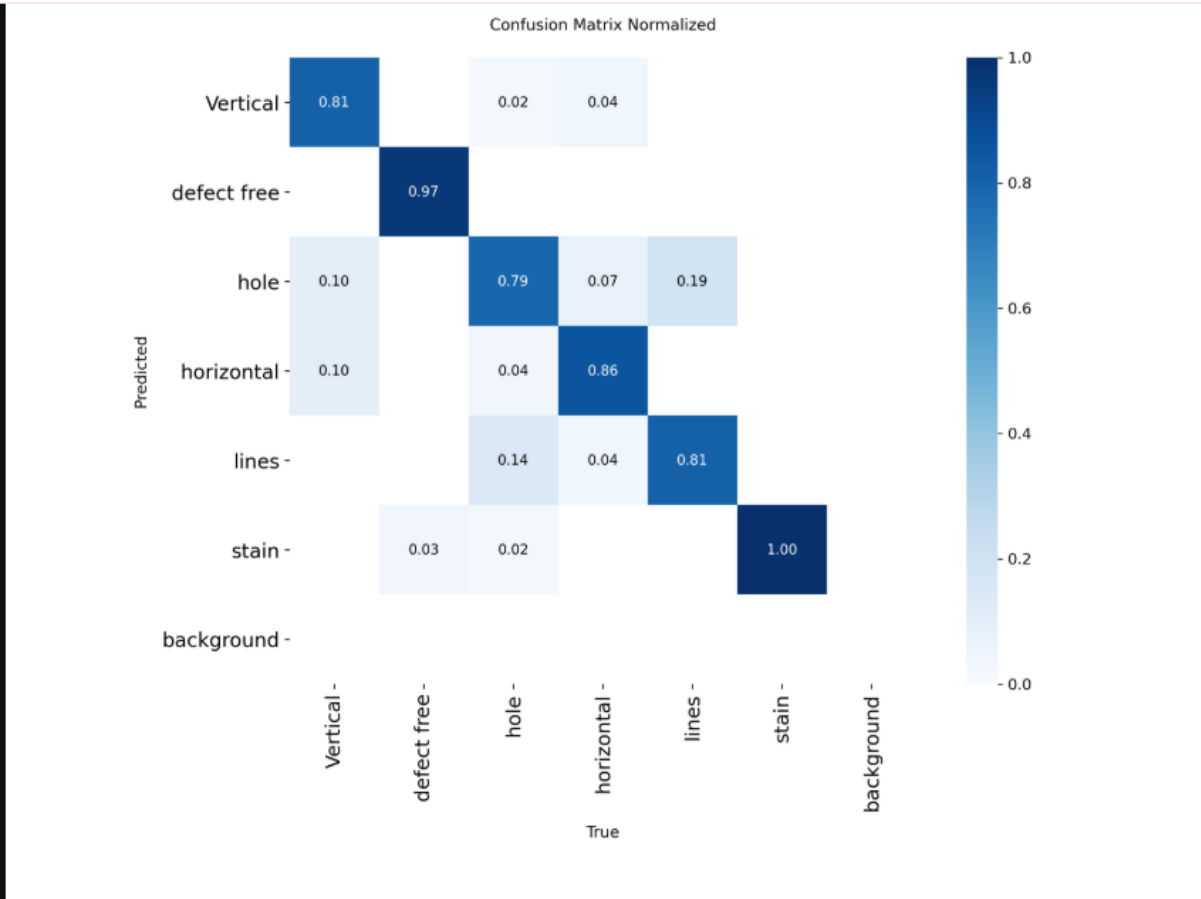
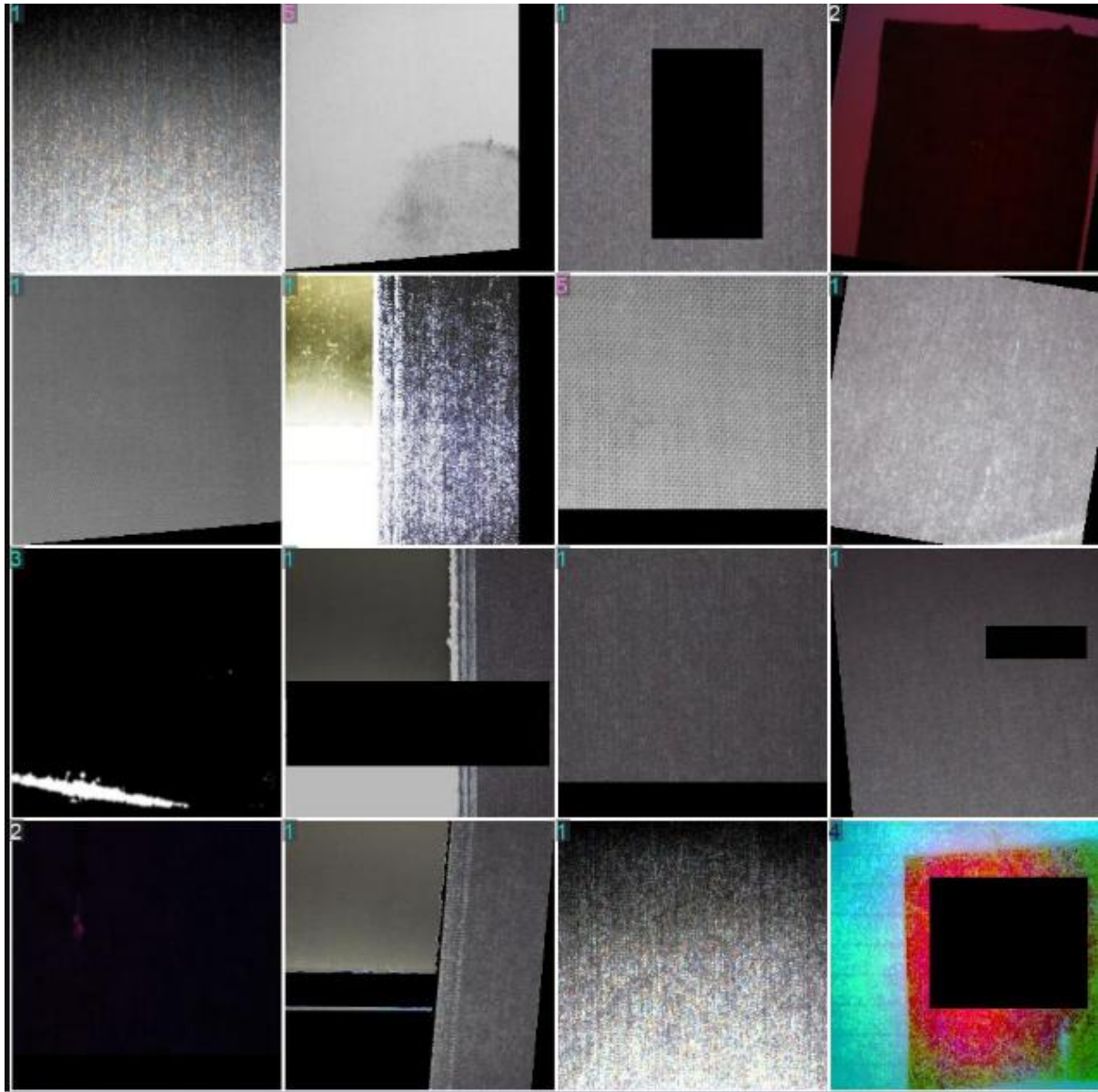


Figure 3: Normalised Confusion Matrix — YOLOv8n Classification on 552-image validation set

7.5 Validation Results

Figure 4 presents the validation batch results illustrating the model's predictions on unseen fabric samples across all six categories.





*Figure 4: Validation batch test results — predicted class labels with confidence overlaid on fabric images*

## 8. FABRIC INSPECTION INFERENCE PIPELINE

### 8.1 Overview

A Fabric Inspection Inference Pipeline was developed using Python and the YOLOv8 Classification framework. The trained model (best.pt) is loaded and used to classify incoming fabric images through a user-friendly graphical interface. The pipeline provides defect classification, confidence estimation, result visualisation, and inspection record generation.

### 8.2 Inference Workflow

8. Image Acquisition — Fabric images selected via graphical file-selection interface. Accepted formats: JPG, JPEG, PNG, BMP.
9. Image Preprocessing — Images resized to 224×224 px internally by the YOLOv8 classification model.
10. Defect Classification — Model predicts one of six categories: Vertical Defect, Horizontal Defect, Hole, Lines, Stain, or Defect-Free Fabric.
11. Confidence Estimation — Probability scores computed for all classes; highest-confidence class selected as final prediction.
12. Result Visualisation — Predicted defect category and confidence score displayed; annotated output image saved.
13. Logging & Record Generation — Defect category, confidence score, inspection time, and image details recorded.

### 8.3 Inference Results — Sample 1 (Stain)

The first inference test was conducted on a fabric image containing a stain defect. The model correctly predicted "stain" with 65.70% confidence.

```
===== FABRIC DEFECT RESULT =====  
Image       : C:/Users/Admin/Downloads/image_1.jpg  
Predicted Class : stain  
Confidence   : 65.70%  
=====
```

All Class Probabilities:  
Vertical : 0.55%  
defect free : 1.24%  
hole : 30.47%  
horizontal : 0.46%  
lines : 1.58%  
stain : 65.70%

PS C:\Users\Admin\Downloads>

Figure 5: Console output — Inference Result 1 (Stain, 65.70% confidence)



Figure 6: Annotated output — Stain defect detected at 65.70% confidence

#### 8.4 Inference Results — Sample 2 (Defect-Free)

The second inference test was performed on a defect-free fabric. The model correctly predicted "defect free" with 64.70% confidence.



```
===== FABRIC DEFECT RESULT =====  
Predicted Class : defect free  
Confidence      : 64.70%  
=====
```

```
All Class Probabilities:  
Vertical : 0.00%  
defect free : 64.70%  
hole : 0.01%  
horizontal : 0.01%  
lines : 0.01%  
stain : 35.27%
```

Figure 7: Console output — Inference Result 2 (Defect-Free, 64.70% confidence)

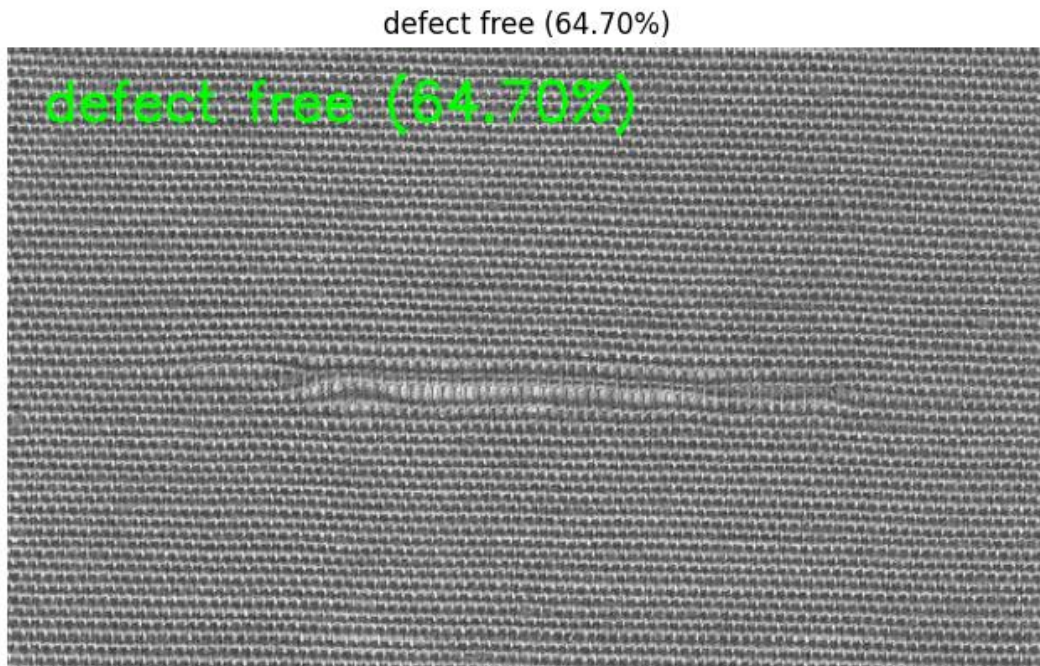


Figure 8: Annotated output — Fabric correctly classified as Defect-Free (64.70%)

### 8.5 Inference Results — Sample 3 (Stain, High Confidence)

A third inference test on a clearly stained fabric sample yielded a high-confidence prediction of 96.07%, validating the model's ability to distinguish stain defects reliably.

```

===== FABRIC DEFECT RESULT =====
Predicted Class : stain
Confidence      : 96.07%
=====

All Class Probabilities:
Vertical : 0.04%
defect free : 2.92%
hole : 0.84%
horizontal : 0.01%
lines : 0.12%
stain : 96.07%

```

Figure 9: Console output — Inference Result 3 (Stain, 96.07% confidence)

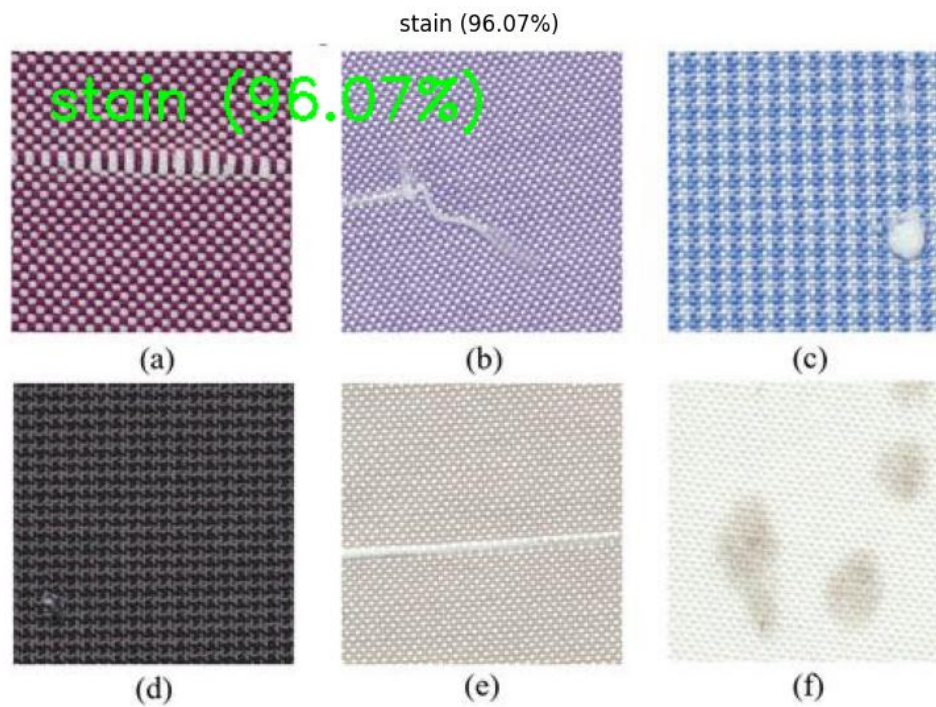


Figure 10: Sample batch — YOLOv8 classification on multiple fabric types showing Stain prediction at 96.07%

## 9. METER POSITION RECORDING AND BATCH LOGGING

### 9.1 Meter Position Recording

In an industrial line-scan fabric inspection system, each captured image corresponds to a specific position on the moving fabric roll. The pipeline associates each image with a meter position value obtained from an encoder or production counter.

| Meter Position (m) | Defect Type       | Confidence |
|--------------------|-------------------|------------|
| 12.5               | Hole              | 96.7%      |
| 18.2               | Stain             | 94.3%      |
| 24.8               | Horizontal Defect | 92.5%      |

This enables defect localisation along the fabric roll and simplifies quality control operations.

### 9.2 Batch-Level Logging

Each inspection entry contains: Image Name, Predicted Class, Confidence Score, Meter Position, and Date/Time. Sample log:

|                |                    |        |         |                     |
|----------------|--------------------|--------|---------|---------------------|
| Image_001.jpg, | Hole,              | 96.7%, | 12.5 m, | 2026-06-10 10:15:24 |
| Image_002.jpg, | Stain,             | 94.3%, | 18.2 m, | 2026-06-10 10:15:31 |
| Image_003.jpg, | Horizontal Defect, | 92.5%, | 24.8 m, | 2026-06-10 10:15:38 |

Logs are exported to CSV files for further analysis and Power BI reporting.

## 10. CONCLUSION

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This report presents a complete automated fabric defect detection and classification system. The detection pipeline — powered by YOLOv11 across 20 defect classes — enables fine-grained bounding-box and polygon-level defect localisation from a dataset of 846 images. The classification pipeline — powered by YOLOv8n across 6 categories — achieves 93.48% Top-1 accuracy on a 552-image validation set after just 10 training epochs, with 100% Top-5 accuracy from epoch 2 onward.

The integrated inference pipeline provides meter-position logging, batch-level CSV export, and Power BI-ready structured output, making it suitable for direct deployment in industrial textile quality control environments.

Future work includes extending the classification head to cover all 20 detection classes, applying transfer learning with larger backbones (YOLOv8m/l), and integrating real-time alerts into shop-floor SCADA systems.

## REFERENCES

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